AGENT EVALUATION
SCIENCE

Technical Companion: Secondary Score-Population and Recovery Checks

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Optional supporting analyses for the run-selection audit. This companion is not a separate AES contribution or an alternative submission. Historical linkage, inherited-grade and simulation boundaries remain explicit.

1 Supplementary aggregate score-column audit

The original aggregate check and later session-grade check are separate evidence tiers; neither establishes a shared benchmark task roster. We inspect every row of the pinned Open Agent Leaderboard release (Open Agent Leaderboard maintainers, 2026): five configurations, five models and six benchmarks, with 150 distinct keys. The artifact preserves aggregate fields, original row order, source hashes, arithmetic and every exception. The Parquet hash is verified against an independent download of the pinned official file. The checked source commit predates the dataset timestamp; the exact generating commit is not recorded.

Establish metric identity before comparing denominators. The reviewed implementation averages non-null numeric session scores for `average_score`, while its orchestrator supplies execution-`COMPLETED` sessions to benchmark aggregation. For SWE-bench, numeric score denotes issue resolution; `successful_sessions` instead counts sessions without a harness error. AppWorld's numeric average uses partial test completion while its primary benchmark score uses task-goal completion, so its 22 differing rows are not denominator evidence. Tau2 metric/grade coverage is not established here. The recorded, completed and planned counts are distinct. Eleven BrowseComp+ rows have positive planned/recorded counts but zero completed/incomplete/missing status counts. Running counts are unreported; status gaps are not proven accounting contradictions. Zero `missing_sessions` in all 150 rows does not establish grade coverage.

Benchmark (25 rows each)	Different score columns	Positive incomplete	Planned $\neq$ recorded	Status coverage gaps
AppWorld	22	11	0	0
BrowseComp+	0	6	0	11
SWE-bench	8	8	4	0
Tau2 airline	0	0	0	0
Tau2 retail	1	4	3	0
Tau2 telecom	0	2	1	0

Table 1 | All-row descriptive checks. Different metrics, incomplete execution and unavailable grades must not be conflated. No cross-benchmark pooling is performed.

Conditional arithmetic and exceptions. Let a, b be the released average and benchmark scores, and T, C the recorded and completed counts. Under complete numeric-grade coverage, aT is an implied resolved total; the reviewed SWE aggregator supports bC. We retain SWE rows with positive counts, equal integral implied totals within 10^{-8}, and separately restrict to planned $= T$. This gives 23/25 rows and 43 within-model configuration pairs, with six strict score-column ordering reversals; the planned-equals-recorded restriction gives 19 rows, 33 pairs and four reversals, all involving DeepSeek Solo. Neither analysis has a tie change. ReAct and ReAct+Shortlisting have identical SWE aggregates; these are configuration comparisons, not independent samples. Public aggregates lack task identities, numeric-grade counts and outcome vectors. Equal products establish conditional arithmetic consistency, not actual equal numerators or common task support.

Score-column illustration and exceptions.

DeepSeek configuration	Recorded	Completed	Average score (%)	Benchmark score (%)
OpenAI Solo	100	38	28.00	73.68
Claude Code	100	100	64.00	64.00
Smolagent	100	100	56.00	56.00
ReAct	100	96	66.00	68.75
ReAct + Shortlisting	100	96	66.00	68.75

Table 2 | Released columns change Solo from lowest to highest among these five configurations. All planned counts are 100. Solo's implied resolved total is 28 under numeric coverage; its `successful_sessions`=38 is a harness property.

The six SWE rows with differing scores and equal implied totals are Claude Code/Kimi (51 over 99 versus 98), Solo/DeepSeek (28 over 100 versus 38), Solo/Kimi (55 over 100 versus 97), Smolagent/Kimi (53 over 93 versus 92), and ReAct and ReAct+Shortlisting/DeepSeek (66 over 100 versus 96). Two other differing SWE rows, ReAct and ReAct+Shortlisting/GPT-5.2, imply 58 versus 57 at identical counts 100 and are excluded from denominator attribution. Claude Code/GPT-5.2/Tau2 retail likewise differs at identical counts. No planned-session zero assignment, grade recovery, corrected leaderboard or prevalence claim follows. This is a reproducible score-column sensitivity check; exact denominator-only attribution needs additional source evidence.

1.1 Direct session-grade corroboration

A separately pinned public Exgentic v2 release (Exgentic, 2026) supplies numeric grades, execution statuses and run/session IDs. We inspect all nine shards (231,705,307 bytes), verify their hashes, and find 10,056 unique sessions; the card states 10,057 and excludes sessions without surviving chat spans. Directed checks of three configurations each recover 100 unique sessions and 100 binary grades:

Archived SWE configuration	All sessions	Reviewed completed status	Error sessions
Solo / DeepSeek	28/100	28/38	62
Solo / Kimi	55/100	55/97	3
ReAct / DeepSeek	66/100	66/96	4

Table 3 | Actual within-run grade sums on two declared populations; every excluded grade is zero. Status success denotes error-free harness execution, not issue resolution.

The source-reviewed completed mapping excludes error/cancelled/unknown, retaining unfinished and limit-reached outcomes. Solo/DeepSeek has only success/error statuses; its 38 success-status sessions include ten unresolved zero grades. The other runs include unfinished/unsuccessful/limit-reached statuses. All six means, completed/recorded counts, steps and costs match the separate historical aggregates. Collection timestamps precede that aggregate's modification timestamp. These matches support association, but the aggregate supplies no run ID or session roster, and the exact exporter/generating commit is unavailable. The v2 schema lacks benchmark task IDs; its full-source endpoint returned 401. Claude/DeepSeek has only 99 v2 sessions versus the aggregate's 100, so its historical score and rank reversal remain unverified. Do not promote the 23 aggregate rows or all historical reversals to direct outcome evidence. The public package contains own replay code, run-level summaries and pins, not source shards or session rows. These source grades are inherited; no independent issue regrading, corrected leaderboard or prevalence claim follows.

2 External task-level reporting transfer: HAL

We adapt two pinned public archived HAL CORE-Bench runs (Kapoor et al., 2025) to the same generic metadata checker and reporter: 90 run/task records over the same 45 task IDs. Source hashes, download/decryption adapter, joins and derived metadata are released; raw traces, answers and archives are excluded. The dataset has no license/card metadata in the inspected repository. The selected runs are small historical artifacts, not a representative HAL sample; membership in the current leaderboard or analyzed paper runs is not established.

Archived run	Tasks	End-to-end successes	Observed exception tasks
HAL Generalist (May 2025)	45	0	45
Claude Code / Opus 4.1 (Dec. 2025)	45	19	Unknown

Table 4 | Recorded task outcomes and exception coverage. Dates, models and scaffolds differ; this is a portability check, not a controlled capability comparison.

The Generalist archive has 45 task-mapped exception records among 57 calls: 33 provider-credit errors and twelve client message-type errors. Every task's raw evaluation reports a JSON parse error with zero correct answers and positive question counts. At the run's recorded evaluator commit, the linked source-semantics table identifies the JSON parsing and exception branches assigning zero correct answers; task success requires all task questions correct. The reproduced 0/45 is therefore known end-to-end failure under that grading rule, not missing grades or proof of a wrong aggregate. It does not identify performance after error-free execution.

The existing reporter reconstructs 0/45 and 19/45 without imputation. Under an explicit no-parse-error or no-call-exception inclusion rule, the Generalist's population is empty and its mean is null/undefined, never zero percent. The processed Claude Code archive lacks status telemetry: its fields stay unknown rather than clean. The report distinguishes 45 known-ineligible Generalist tasks from 45 unknown-eligibility controls, including per-rule null/absent counts. The checker validates all 90 identities/grades; no task IDs or errors are invented. This is a computational cross-corpus transfer, not independent human adoption or a new failure-aware evaluation method.

The public trace dataset is pinned to `e7dcedc82b4f4bc819a170fd6616bdb44841c71e`. Its Generalist run records evaluator commit `db2a824aa6a07ce8f1097258327fdfb7dd04c2ac`. Artifact manifests pin two archive hashes and the official publicly documented decryption code. No new agent calls, model grading or human participants are required.

2.1 Size-selected mixed-status archives

To test nonempty eligible and ineligible populations, we fixed an archive-size selection plan before inspecting new scores: select the three smallest full CORE-Bench Generalist archives at the same dataset revision, excluding slim exports and the previously inspected all-error archive. The inventory records 32 candidates. No candidate fit the initial 10 MB budget; the declared three-archive alternative totals 96,494,907 bytes. Every selected archive is reported, including the run without recorded errors. This local prescore plan is not external preregistration; archive size is not representative sampling.

The same checker and reporter reproduce all 135 unique task/run grades on an identical 45-task universe. All 1,965 archived calls join to that universe: 458 V3 calls, 546 R1 and 961 Sonnet. V3 has one task with an API exception; the other runs have none among archived calls. Sonnet nevertheless has 40 recorded evaluator parse errors. Absence of an archived call exception therefore differs from valid parsing and does not prove execution completion.

The three run-recorded evaluator commits (`7e56e668`, `c7354ebd`, `9858be97`) have byte-identical CORE-Bench evaluator files. Their parsing-exception branch assigns zero correct answers;

Historical Generalist run	All-task score	Parse errors	No-parse-error score
DeepSeek V3-0324 (Aug. 2025)	4/45	1	4/44
DeepSeek R1 (Sept. 2025)	1/45	0	1/45
Claude 3.7 Sonnet (June 2025)	4/45	40	4/5

Table 5 | All selected archives, including the unchanged case. Filtering creates a conditional ordering from an all-task tie; it is not a rank reversal or corrected leaderboard score.

success requires every task question correct. Every parse-error task has a known zero grade, positive question counts and the same recorded JSON parsing error. No grade is imputed or recovered. No-error status means absence of the evaluator's error field under this pinned implementation, not successful scientific reproduction.

The two 4/45 end-to-end scores are both 8.89%, while their no-parse-error means are 9.09% and 80%. These different populations explain conditional score sensitivity; neither conditional mean estimates performance after fixing errors. Dates, models, budgets and configurations differ. The released adapter, plan, archive/source hashes and derived metadata support raw replay without new inference; raw traces remain excluded. This extends the reporter beyond the original empty-support boundary while preserving the original grade policy.

3 Global recorded-winner disclosure

For an equally weighted common roster of N tasks, grades in $[0, 1]$, independent unrestricted remaining grades and unit cost per disclosed model/task grade, seek a strict certificate $L_w - U_m > \tau$ for every rival m, with $\tau = 10^{-12}$. Disclosing winner grade y raises its lower bound by y/N; disclosing rival grade y lowers its upper bound by $(1 - y)/N$. For each model sort its corresponding omitted gains decreasing. Let $A_w(k)$ be the sum of the largest k winner gains and $B_m(q)$ the largest q rival gains. Define

$$q_m(k) = \min\{q : L_w + A_w(k) - U_m + B_m(q) > \tau\}, \qquad C^* = \min_{k:\ \text{all } q_m(k) \text{ exist}} \left\lceil k + \sum_{m \neq w} q_m(k) \right\rceil .$$

Exact offline optimality. For fixed k, replacing a smaller winner disclosure by a larger one weakly relaxes every rival constraint. Each rival's largest gains maximize its upper-bound reduction at any count; its first feasible prefix is minimal. Rival sets are disjoint, so those minima are simultaneously feasible and additive. Enumerating k gives the global optimum. A tied recorded maximizer cannot have a strict certificate. This is outcome-informed verification using established uncertainty-query reasoning (Olston and Widom, 2000; Gupta et al., 2011), not a new general theory or prospective budget.

Fixed blinded selector. Choose the largest lower-bound incumbent and largest upper-bound rival; stop when their gap exceeds τ. Otherwise query the one of these two models with the wider interval (more missing grades), then its lexicographically first unavailable task. Model ties are lexical. The selector receives revealed grades and missing identities only; a separate retrieval step supplies one withheld existing grade. No policy or parameter search was performed and no competitive guarantee is asserted.

Mask	Restore every grade	Offline optimum	Fixed selector
Assistant ending	54	5	5
Native stop	175	60	76
Exporter completed	33	2	2

Table 6 | Disclosure counts under all existing masks; all certify the released-roster GPT-5.6 Sol mean. Exporter covers ten models, the other masks twelve. No grade is newly generated.

The winner has no omitted positive grades: native-stop and exporter each omit one zero grade. Its initial lower bound already equals its full recorded mean, and neither procedure queries its grades, limiting empirical complexity. Exhaustive 3-model/2-task binary fixtures cover all 4,096 grade/mask combinations, including 1,584 strict-winner cases with missing winner grades; another 150 fractional fixtures compare oracle counts against all disclosure subsets. Every actual query trace and final certificate is checked. An independent checker reproduced release counts and found a near-tolerance binary-rounding inconsistency; the corrected implementation uses exact rational interpretations of canonical JSON decimals for feasibility, policy and validation, with floats only for display. Regression checks cover equality and just-above-tolerance gaps. These are simulated disclosures of available grades, not natural recovery costs, measured deployment benefit or achievement validation.

3.1 Outcome-independent masking stress test

The original completion masks do not hide positive winner grades. To exercise that harder case without selecting masks by outcomes, we fixed three omission probabilities (0.10, 0.25, 0.50) and thirty deterministic seeds per probability before generating masks. Every one of the 720 model/task cells is withheld independently; no mask is rejected or resampled. This extension was developed after the earlier analyses: the local plan is not external preregistration or independent confirmation. Its explanatory amendment records timing; sampling and comparison choices are unchanged.

The existing selector is compared on identical masks with two grade-blind retrieval orders: lexicographically sorted model/task identities and a separately seeded random permutation. Both stop at the first strict interval certificate. Restore-all retrieves every omitted grade; the offline oracle uses hidden truth. All methods target the same equally weighted full-roster mean, use exact-decimal bounds and count unit-cost retrievals of existing grades. All ninety cases remain in the denominator, including any initial certificate or full-mean tie; this matrix has a unique strict winner and no initial certificates.

Omission rate	Selector	Lexical	Random	Offline oracle	Restore all
10%	9.87	20.40	50.00	6.13	70.37
25%	53.73	133.77	159.90	27.83	180.13
50%	208.50	339.40	339.57	99.00	361.07

Table 7 | Mean disclosures over all thirty seeds per rate. Every mask omits at least one positive winner grade; every method certifies the recorded winner in every case. The oracle is outcome-informed, not a prospective budget.

The selector uses fewer queries than random retrieval in 90/90 cases and than lexical retrieval in 89/90, with one tie. Its mean cost exceeds the oracle at every rate; at 50% omission it is 208.50 versus 99.00. The artifact provides all-rate cost ranges, quantiles, paired differences and omitted-winner-grade counts, checks reconstructed certificates against exact full means, and repeats the experiment deterministically. An independent checker reproduces the aggregate calculations. Artificial independent missingness on the same roster does not validate performance under actual missingness, model generalization, competitive guarantees, adoption or achievement. No new inference, grading or participants are involved.

3.2 Interval-aware controls and archived-corpus transfer

Lexical and random traversal can waste queries on eliminated models. We add two fixed interval-aware controls without tuning the original policy or masks. Active-candidate traversal queries the lexically first missing task of the lexically first model with an unrevealed grade whose upper bound is at least the largest lower bound minus τ. Challenger-focused retrieval queries the largest-upper rival before the largest-lower incumbent, falling back to the incumbent if the rival is fully known. Both check the same strict certificate before retrieving, and see known grades and missing identities only.

The local plan retains all ninety original masks and additionally applies the same rates/seeds to two recorded HAL CORE-Bench rosters: all five previously inspected archives on 45 common task IDs, and the three size-selected mixed archives. These two HAL rosters overlap. Archive identity distinguishes configurations; dates, models and scaffolds differ, so the target is the archived-run recorded mean, not a controlled capability comparison. This external-corpus transfer uses previously inspected data, not an untouched holdout or independent achievement validation. One control archive's grades derive from archived success/failure labels; other grades follow the recorded evaluator convention.

Roster/omission	Original	Active	Challenger	Oracle	Restore all
Wild: 10%	9.87	15.03	11.13	6.13	70.37
Wild: 25%	53.73	57.37	59.90	27.83	180.13
Wild: 50%	208.50	202.23	222.80	99.00	361.07
HAL five: 10%	0.00	0.00	0.00	0.00	21.50
HAL five: 25%	4.63	5.97	4.87	2.90	57.20
HAL five: 50%	50.30	44.50	63.67	30.73	110.13

Table 8 | Mean queries over thirty masks per row; every listed method certifies the strict recorded winner in all cases. Active traversal beats the original policy's mean at 50% omission in both corpora. Exact original lexical/random results are preserved in the artifact.

At 50% omission active traversal beats the original policy in 21/30 Wild and 27/30 HAL cases. All thirty HAL five-run 10% masks initially certify, so zero queries demonstrate initial sufficiency, not retrieval skill. The three-run HAL roster has two tied recorded maximizers: all ninety masks remain uncertified and every offline oracle is null. Query costs on those unresolved cases do not represent successful recovery. No roster or tie case is discarded.

The artifact reports all 270 cases and seven methods, including lexical/random costs, paired differences, mask hashes and per-case certificate flags without task contents or grades. Two complete replays match; each final interval certificate is independently reconstructed, and a separate checker reproduces all mask hashes and costs. The original ninety-mask results are unchanged. These stronger controls reveal policy-dependent costs, not a new competitive guarantee, naturally observed retrieval savings or a reason to replace available grades with a filtered score.

References

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